

Order Flow Imbalance and Short-Horizon FX Returns: A Multiple-Testing-Corrected Double Machine Learning Analysis

Charles Glah

Founder, DeepAlgo Sovereign

21 August 2026

ABSTRACT

Order flow imbalance (OFI) has a long lineage in market microstructure as a candidate driver of short-horizon price impact, from Kyle's (1985) linear price-impact model through the bulk volume classification tradition underlying VPIN (Easley, López de Prado & O'Hara, 2012). Practitioner claims that a given order-flow construction "predicts" returns are common but are rarely subjected to a causal standard that separates a structural relationship from confounding by time-of-day, volatility clustering, and recent momentum — nor to correction for the fact that testing many instruments and time partitions inflates the chance of a false positive. This note applies Double Machine Learning (Chernozhukov et al., 2018) to a transparent, bar-level OFI proxy across a systematic 32-instrument, 730,013-observation sweep of 30-minute FX, metals, and sovereign-bond-CFD data, and subjects the resulting 256-test battery to Benjamini-Hochberg false discovery rate correction. Under the correction family appropriate to a single pre-specified sweep, no instrument-session pairing retains a statistically defensible relationship between OFI and 90-minute forward returns. A small number of nominally significant results survive a looser, per-instrument correction, but carry effect sizes too small to be economically meaningful. The result is reported as a genuine null: a signal that would appear promising under the single-pair, uncorrected screen many practitioner claims rely on does not survive a stricter, causally motivated, multiple-testing-aware test.

AI disclosure statement. This working paper was prepared with the assistance of an AI-based analysis tool, used for editorial review and independent verification of the statistical results reported. The underlying data, code, and empirical analysis are the author's own; every figure in this paper was independently re-derived and cross-checked against the source data — including via a second, independent statistical library — before publication.

1. Introduction

A recurring claim in quantitative trading practice is that some measure of order flow imbalance — the net directional pressure implied by recent trading activity — has predictive power for near-term price movement. The theoretical basis is old and well established: Kyle (1985) models price impact as linear in signed order flow, and the informed-trading literature that follows Easley, López de Prado and O'Hara (2012) treats volume-synchronised measures of order-flow toxicity as a genuine structural feature of markets, not an artefact.

The practical difficulty is that raw correlations between an order-flow proxy and subsequent returns are easy to confound. FX markets exhibit strong, well-documented session effects (liquidity, volatility, and typical order sizes differ systematically between the Asian, London and New York windows), volatility clustering, and short-horizon momentum — any of which can produce a spurious association between an OFI measure and forward returns without the OFI measure itself carrying any causal information. A raw predictive correlation that is in fact driven by, say, the London-session volatility regime is not a tradable causal signal; it is a restatement of the session effect.

This note asks a narrow, falsifiable question: does a simple, fully disclosed OFI construction retain predictive power for short-horizon FX returns once session, volatility, and momentum confounders are properly controlled for — and does that relationship survive when tested systematically across a broad instrument universe, with the resulting multiplicity of tests properly corrected for?

2. Data

Thirty-minute (M30) mid-price candles were retrieved from the OANDA REST API for 32 instruments spanning FX majors and crosses, two precious metals, and two sovereign-bond CFD proxies:

```
AUD_CAD, AUD_CHF, AUD_JPY, AUD_NZD, AUD_USD, CAD_CHF, CAD_JPY, CHF_JPY, DE10YB_EUR,  
EUR_AUD, EUR_CAD, EUR_CHF, EUR_GBP, EUR_JPY, EUR_NZD, EUR_USD, GBP_AUD, GBP_CAD, GBP_CHF,  
GBP_JPY, GBP_NZD, GBP_USD, NZD_CAD, NZD_CHF, NZD_JPY, NZD_USD, USB10Y_USD, USD_CAD,  
USD_CHF, USD_JPY, XAG_USD, XAU_USD
```

The confirmed retrieval window, taken from execution logs rather than assumed from configuration, runs from **26 July 2024 to 8 June 2026**. After removing incomplete candles and rows with missing confounder values, per-instrument sample sizes range from 18,885 to 23,117 bars, for a combined **730,013 full-sample observations** across the 32-instrument sweep.

Three trading-session partitions were defined by UTC hour: Asian (00:00–07:59), London (07:00–15:59), New York (13:00–21:59). These windows overlap by construction, reflecting genuine liquidity-handover periods rather than a strict partition of the day; two hours (22:00–23:59 UTC) fall outside all three named sessions and are represented only in the full-sample slice. This is disclosed here rather than left implicit, as it affects how the session-level results should be read.

3. Methodology

3.1 *Order flow imbalance construction*

OANDA's retail data feed does not expose limit-order-book depth or individual signed trade prints, so OFI here is necessarily a bar-level proxy rather than literal order-book flow. For each 30-minute bar, a directional-pressure measure is computed as the signed intrabar move normalised by the bar's own range, then scaled by a log-volume weight:

$$\begin{aligned}\text{OFI}_{\text{raw}} &= (\text{close} - \text{open}) / (\text{high} - \text{low}) \\ \text{OFI} &= \text{OFI}_{\text{raw}} \times \ln(1 + \text{volume})\end{aligned}$$

This construction sits in the same conceptual family as bulk volume classification (the methodology underlying VPIN): it infers net directional pressure from bar-level price and volume rather than from individually signed trades. It is disclosed as a proxy, not presented as ground-truth order flow.

3.2 *Confounder set*

Fifteen confounders were included in both nuisance models: 20-period realised volatility, 14-period ATR, a binary trend-direction flag (sign of close relative to a 20-period EMA), 20-period cumulative log-return momentum, hour-of-day, day-of-week, a binary flag for UTC hours coinciding with recurring macro data releases, and four lagged values each of OFI and of the raw log return (lags of 1, 2, 3 and 5 bars).

3.3 *Double Machine Learning*

The Double Machine Learning partialling-out estimator (Chernozhukov et al., 2018) was used to isolate the component of the OFI–return relationship not attributable to the observed confounders. Two nuisance models — gradient-boosted regression trees (200 estimators, maximum depth 3, learning rate 0.05, subsample fraction 0.8) — are fit via 5-fold cross-fitting to predict, respectively, the treatment (OFI) and the outcome (log return over the following three bars, a 90-minute horizon) from the confounder set. Cross-fitting folds are contiguous, non-shuffled blocks in chronological order, rather than randomly

shuffled folds. The **structural IC** is the Pearson correlation between the two sets of out-of-fold residuals: any component of the raw OFI–return correlation that is fully explained by session timing, recent volatility, or momentum is differenced out before this correlation is computed.

3.4 Multiple testing correction

The sweep produces $32 \text{ instruments} \times 4 \text{ slices (full sample plus three sessions)} = 128$ tests per estimator (raw correlation, structural/DML correlation), 256 tests in total. Benjamini-Hochberg false discovery rate correction (Benjamini & Hochberg, 1995) was applied at two levels of granularity, reported separately rather than selecting whichever is more favourable: (i) **pooled** across the full 128-test family per estimator, and across all 256 tests combined — the correction appropriate to what this sweep actually is, a single pre-specified systematic test, not 32 independently designed studies; and (ii) **per-instrument**, correcting only within each instrument's own four-slice family, for comparison. Reporting both is itself the point: the second, looser framing is a common way an honest multiplicity problem gets understated in practice, and the gap between the two results below is informative in its own right.

4. Results

Effect sizes across all 128 tests were small throughout: median $|\text{raw IC}| = 0.0090$ (range 0.0001–0.0353); median $|\text{structural IC}| = 0.0105$ (range 0.0000–0.0336). At the largest observed magnitude, $R^2 \approx \text{IC}^2$ implies at most roughly 0.1% of forward-return variance explained — a ceiling that applies before any significance test is even considered.

Uncorrected significance. 15 of 128 raw-correlation tests and 17 of 128 structural (DML) tests were nominally significant at $p < 0.05$. With 128 independent tests, roughly 6.4 false positives are expected by chance alone at this threshold; 15–17 is modestly above that baseline but not compellingly so on its own.

Pooled BH-FDR correction (the appropriate family for this sweep): **zero of 128 tests survive for either estimator**, at both $\alpha = 0.05$ and the more permissive $\alpha = 0.10$. Pooling both estimators into a single 256-test family produces the same result: **zero of 256 survive** at $\alpha = 0.10$.

Per-instrument BH-FDR correction (four tests per instrument, structural IC, $\alpha = 0.10$): seven of 32 instruments retain at least one nominally surviving slice — AUD_CAD (3 of 4 slices), USD_CHF (3 of 4), EUR_GBP (2 of 4), GBP_AUD (2 of 4), EUR_CHF, EUR_NZD, and USB10Y_USD (1 of 4 each). Even among these survivors, structural IC magnitudes range only 0.014–0.034.

CORRECTION	FAMILY SIZE	ESTIMATOR	SURVIVORS

None (nominal $p < 0.05$)	—	Raw	15 / 128
None (nominal $p < 0.05$)	—	Structural (DML)	17 / 128
BH-FDR, $\alpha = 0.10$	Pooled, 128	Raw	0 / 128
BH-FDR, $\alpha = 0.10$	Pooled, 128	Structural (DML)	0 / 128
BH-FDR, $\alpha = 0.10$	Pooled, 256 (both estimators)	Combined	0 / 256
BH-FDR, $\alpha = 0.10$	Per-instrument, 4	Structural (DML)	7 / 32 instruments

5. Discussion

Under the correction family that actually describes this study — one pre-specified, systematically executed sweep across a fixed universe, not 32 independently conceived hypotheses — the result is an unambiguous null. This OFI construction, at 30-minute granularity and a 90-minute forward horizon, does not exhibit a causally robust relationship with short-horizon FX returns once standard microstructure confounders are controlled for.

This is reported as a genuine finding, not a failed one. The value of the exercise is precisely that it demonstrates how a signal can appear promising under the kind of single-pair, uncorrected raw-correlation screen that much practitioner and even some published work relies on — 15–17 nominally significant results out of 128 tests, or a locally impressive per-instrument result for AUD_CAD or USD_CHF — and fail to survive the combination of causal adjustment and correction for the true scope of the search. The gap between the per-instrument and pooled correction results in Table 1 is itself the more general lesson: the choice of correction family is not a technicality, it materially changes which claims a study can support.

6. Limitations

- **OFI is a bar-level OHLCV proxy, not order-book flow.** OANDA's retail feed provides no L2 depth or individually signed trades. A genuine order-book-derived OFI measure (e.g. from an exchange feed or a prime-broker tape) could behave differently; this result should not be read as a statement about order-flow signals in general, only about this specific, fully disclosed construction.
- **Single fixed horizon and granularity.** Only a 90-minute forward horizon at 30-minute granularity was tested. A relationship invisible at this horizon could exist at another; a horizon sweep is a natural extension, though one that would itself need to correct for the additional multiplicity it introduces.

- **Session partitions overlap and do not cover the full day**, as noted in Section 2 — a labelling property of the session definitions, not a data gap, but one that affects how the by-session rows should be interpreted.
- **Cross-fitting folds are contiguous but not purged or embargoed.** Because the outcome is a forward return spanning multiple bars, observations near a fold boundary can have overlapping outcome windows across the train/test split. A purged-and-embargoed cross-fitting scheme (López de Prado, 2018) — which explicitly removes training observations whose own label window overlaps the held-out span — is a stricter and more defensible variant than the contiguous-block cross-fitting used here, and is a natural next check on this result.
- **Nuisance model specification was fixed, not tuned**, across all 32 instruments. DML's asymptotic guarantees depend on the nuisance models achieving adequate predictive accuracy; a single untuned hyperparameter configuration applied uniformly is a conservative but not exhaustive test of that assumption.

7. Conclusion

Across a systematic, pre-specified sweep of 32 instruments and 730,013 half-hourly observations, no order-flow-imbalance–return relationship survives properly corrected multiple testing, and the small number of nominally significant results that survive a looser per-instrument correction carry economically negligible effect sizes. Reporting this as a clean null, rather than reframing it around whichever subset of results looks strongest, is the more defensible reading of the evidence — and is consistent with the broader point that causal validation, applied honestly, more often rejects a candidate signal than confirms one.

References

- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300.
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1), C1–C68.

Easley, D., López de Prado, M. M., & O'Hara, M. (2012). Flow toxicity and liquidity in a high-frequency world. *Review of Financial Studies*, 25(5), 1457–1493.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6), 1315–1335.

López de Prado, M. M. (2018). *Advances in Financial Machine Learning*. Wiley.